Dataset Details

1. Dataset Name and Description

Dataset: Default of Credit Card Clients (UCI Machine Learning Repository). The dataset contains 30,000 client observations and 25 variables: a customer ID, 23 explanatory variables (X1–X23), and one binary target (Y). It records credit limits, demographic characteristics, six months of repayment status, statement bills, payment amounts, and whether a client defaulted on payment in the following month.

Project purpose: The Banking Loan Risk Analysis Application validates and transforms these records, loads curated customer/credit/bill/payment data into MySQL, and uses them for portfolio analytics, rule-based risk indicators, and dashboards.

2. Dataset Source and Format

Source: UCI Machine Learning Repository — Default of Credit Card Clients (dataset 350).
Link: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients
Original file format: XLS (Excel); the project extractor accepts .xls and .xlsx, reads sheet **Data**, and uses the second spreadsheet row as the header (**header=1**).
Coverage: April–September 2005 repayment/billing/payment history; target indicates default next month.

3. Key Fields and Definitions

Source field	Description
ID	Customer record identifier
X1 / LIMIT_BAL	Amount of given credit (NT dollars)
X2 / SEX	Sex: 1 = male; 2 = female
X3 / EDUCATION	Education: 1 = graduate school; 2 = university; 3 = high school; 4 = others; source also includes codes 0, 5, 6
X4 / MARRIAGE	Marital status: 1 = married; 2 = single; 3 = others; 0 also occurs
X5 / AGE	Age in years
X6–X11 / PAY_0, PAY_2–PAY_6	Repayment status, September through April 2005; positive values indicate delay in months; -1 paid duly; -2 and 0 are present but their exact semantics are not fully documented
X12–X17 / BILL_AMT1–6	Bill statement amounts, September through April 2005 (NT dollars)
X18–X23 / PAY_AMT1–6	Previous payment amounts, September through April 2005 (NT dollars)
Y / default payment next month	Binary outcome: 1 = default; 0 = no default

4. Data Preparation in This Project

The ETL checks the required source columns, validates values and ranges (including age, credit limit, categorical codes, repayment status and target), standardizes column names, converts numeric types, derives demographic labels and age bands, and reshapes monthly bills and payments for relational storage. Invalid records are tracked in data-quality/audit processes. Incremental processing uses record hashes to identify changes.

5. Sample Data (First Eight Records Provided)

The source table has a two-row header: X1–X23 and Y are variable codes, while the second header row contains the corresponding field names. For readability, the same eight provided observations are presented in four linked tables, keyed by ID. Amounts are in NT dollars.

5.1 Customer and Demographic Fields

ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	Y
1	20000	2	2	1	24	1
2	120000	2	2	2	26	1
3	90000	2	2	2	34	0
4	50000	2	2	1	37	0
5	50000	1	2	1	57	0
6	50000	1	1	2	37	0
7	500000	1	1	2	29	0
8	100000	2	2	2	23	0

5.2 Repayment Status Fields

ID	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6
1	2	2	-1	-1	-2	-2
2	-1	2	0	0	0	2
3	0	0	0	0	0	0
4	0	0	0	0	0	0

ID	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6
5	-1	0	-1	0	0	0
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	-1	-1	0	0	-1

5.3 Monthly Bill Amounts

ID	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6
1	3913	3102	689	0	0	0
2	2682	1725	2682	3272	3455	3261
3	29239	14027	13559	14331	14948	15549
4	46990	48233	49291	28314	28959	29547
5	8617	5670	35835	20940	19146	19131
6	64400	57069	57608	19394	19619	20024
7	367965	412023	445007	542653	483003	473944
8	11876	380	601	221	-159	567

5.4 Previous Payment Amounts

ID	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6
1	0	689	0	0	0	0
2	0	1000	1000	1000	0	2000
3	1518	1500	1000	1000	1000	5000
4	2000	2019	1200	1100	1069	1000

ID	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6
5	2000	36681	10000	9000	689	679
6	2500	1815	657	1000	1000	800
7	55000	40000	38000	20239	13750	13770
8	380	601	0	581	1687	1542

Note: These eight sample records are reproduced from the table supplied for this deliverable. The complete source dataset contains 30,000 observations. The Y column is the observed outcome and is distinct from the application's configurable rule-based risk bands.